

- 6.6** Design a four-bit shift register with parallel load using D flip-flops. There are two control inputs: *shift* and *load*. When *shift* = 1, the content of the register is shifted by one position. New data are transferred into the register when *load* = 1 and *shift* = 0. If both control inputs are equal to 0, the content of the register does not change. (HDL—see Problem 6.35(c), (d).)
- 6.10** Design a serial 2's complemeter with a shift register and a flip-flop. The binary number is shifted out from one side and its 2's complement shifted into the other side of the shift register. (HDL—see Problem 6.35(j).)
- 6.16*** The BCD ripple counter shown in Fig. 6.10 has four flip-flops and 16 states, of which only 10 are used. Analyze the circuit and determine the next state for each of the other six unused states. What will happen if a noise signal sends the circuit to one of the unused states? (HDL—see Problem 6.54.)

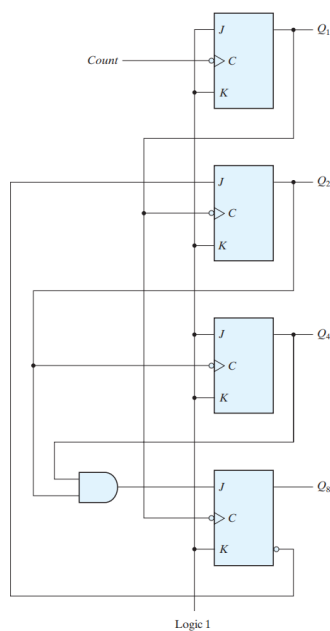
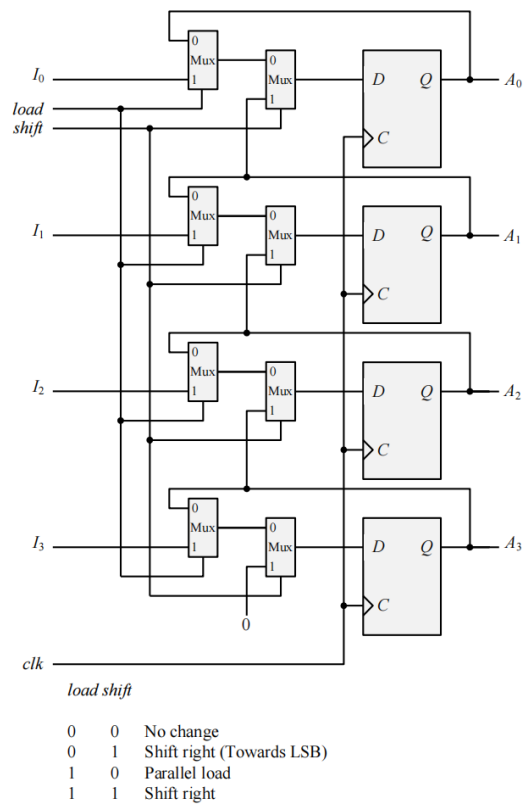


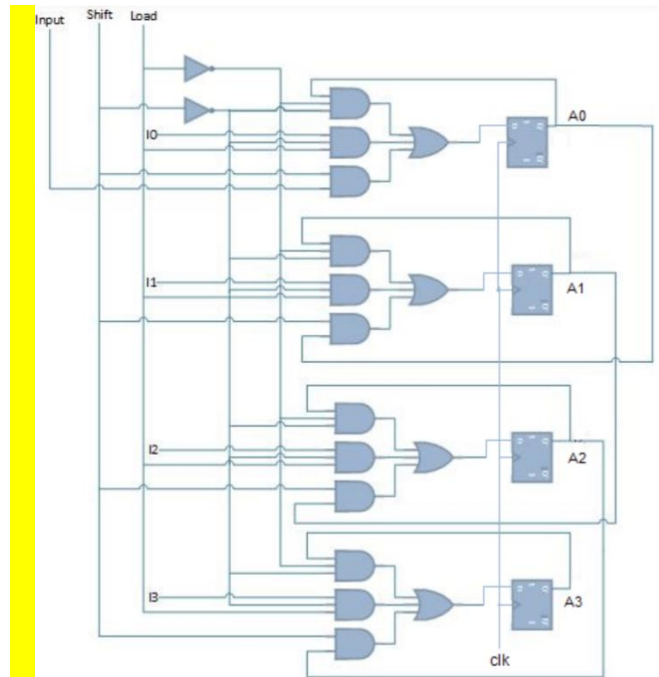
FIGURE 6.10

- 6.19** The flip-flop input equations for a BCD counter using T flip-flops are given in Section 6.4. Obtain the input equations for a BCD counter that uses (a) JK flip-flops and (b)* D flip-flops. Compare the three designs to determine which one is the most efficient.
- 6.27** Using JK flip-flops,
- Design a counter with the following repeated binary sequence: 0, 1, 2, 3, 4, 5, 6. (HDL—see Problem 6.50(a), 6.51.).
 - Draw the logic diagram of the counter.

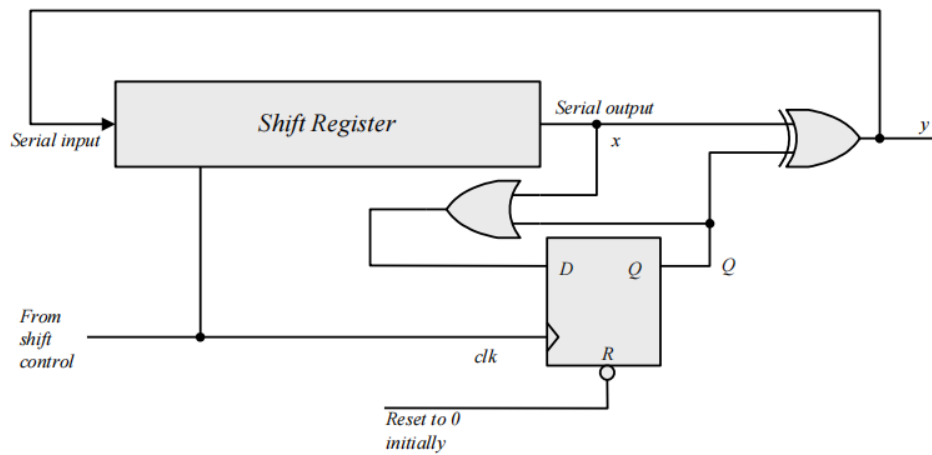
6.6



OR



6.10



6.16

Q8 Q4 Q2 Q1 : 1010 1100 1110 Self correcting
Next state: 1011 1101 1111
Next state: 0100 0100 0000

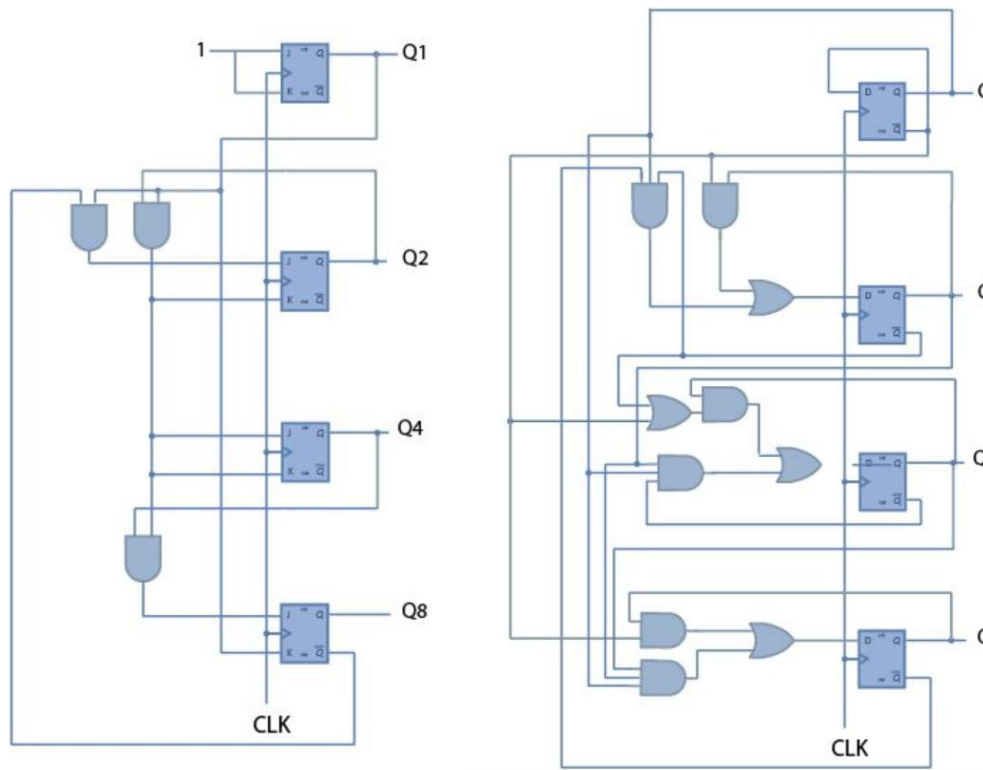
1010 → 1011 → 0100
1100 → 1101 → 0100
1110 → 1111 → 0000

6.19

When comparing designs with T, D and JK flip flops we can notice that the JK design is the easiest and most efficient since the number of gates is three, meaning propagation delay is the smallest.

JK

D



6.27

Present state <i>ABC</i>	Next state <i>ABC</i>	Flip-flop inputs					
		J_A	K_A	J_B	K_B	J_C	K_C
000	001	0	x	0	x	1	x
001	010	0	x	1	x	x	1
010	011	0	x	x	0	1	x
011	100	1	x	x	1	x	1
100	101	x	x	0	0	1	x
101	110	x	0	1	x	x	1
110	000	x	1	x	1	0	x
111	xxx	x	x	x	x	x	x

